

Constraint Before Reference: Operator Semantics, Reification, and Symbolic Conflict

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Abstract

Persistent symbolic and ideological conflicts are often attributed to deep metaphysical or factual disagreement. This paper argues that many such conflicts are more accurately diagnosed as semantic failures arising from a misordering of priority between constraint and reference. In a wide class of linguistic, institutional, and computational systems, meaning functions primarily by constraining admissible futures rather than by denoting objects. When symbols that operate as constraints are reclassified as referential absolutes, negotiable procedural disagreements collapse into non-negotiable existential claims.

The paper develops a constraint-first framework grounded in philosophy of language, operational semantics, and category theory. It formalizes linguistic operators as functions on event histories that irreversibly restrict future possibilities, provides a small-step operational semantics with soundness and completeness results, and lifts this account to a functorial semantics in which symbols correspond to endofunctors on spaces of admissible futures. Reification is shown to constitute a semantic type error with predictable structural consequences, including loss of negotiability and escalation of conflict.

The framework explains the role of semantic compression and drift in long-lived symbolic systems, characterizes narrative as a constraint-preserving encoding, and offers precise diagnostics for distinguishing factual disagreement from operator misclassification. Applications are developed across legal systems, organizational governance, computational architectures, and multi-agent coordination, demonstrating how operator-level reframing can expand resolution space without requiring consensus on beliefs or interpretations.

By restoring category discipline between constraint and reference, the paper provides a neutral, formally explicit account of meaning that clarifies why many disputes persist despite shared normative commitments and identifies structural conditions under which disagreement can remain negotiable rather than absolute.

1 Introduction

Persistent conflicts across legal, political, organizational, and computational domains are frequently described as deep, principled, or metaphysical in nature. They are often attributed to incompatible worldviews, irreconcilable values, or fundamental disagreements about reality itself. Yet in many such cases, closer inspection reveals substantial agreement on practical norms governing action,

responsibility, and coordination. What remains unresolved is not what should be done, but how certain symbols, terms, or roles are understood to function within the system that governs doing.

This paper advances the thesis that a large class of persistent symbolic conflicts arise from a specific semantic failure: the inversion of priority between *constraint* and *reference*. Symbols that originally functioned to regulate admissible actions, identities, or future states are misclassified as denoting absolute objects or facts. Once this misclassification occurs, procedural disagreement is transformed into existential opposition, rendering negotiation structurally impossible.

The central claim is not that reference is unimportant, but that it is secondary. Constraint is logically and operationally prior. Meaning stabilizes action before it stabilizes description.

1.1 A Motivating Case: Jurisdictional Conflict

Consider a recurring form of legal conflict involving jurisdiction. Two courts, agencies, or institutional bodies may dispute which of them has authority over a given case. Such disputes are often framed as disagreements about facts: where an event occurred, which law applies, or which entity possesses legitimate power.

However, jurisdiction does not denote a physical object or discoverable property. It is an operator: a rule that constrains which actions are permitted by whom and under what conditions. When jurisdiction is treated as a referential fact rather than a procedural constraint, disputes escalate. Each party seeks to establish exclusive ownership of an alleged object (true authority), rather than to negotiate the boundaries of permissible action.

In practice, many jurisdictional disputes are resolved not by discovering new facts, but by re-defining procedural constraints: carving out exceptions, establishing precedence rules, or introducing shared oversight mechanisms. The conflict dissolves once the operator structure is restored.

This pattern recurs across domains. The case is representative rather than exceptional.

1.2 Problem Statement

The prevailing semantic frameworks used to analyze disagreement emphasize truth-conditions, reference, and correspondence. While powerful, these frameworks are ill-suited to explaining why conflicts persist in contexts where truth-value disagreement is minimal or irrelevant.

This paper proposes an alternative explanatory axis. It distinguishes symbols that function primarily as *operators* constraining admissible futures from symbols that function primarily as *referents* denoting objects or states of affairs. Conflicts arise when operator symbols are reinterpreted as referential absolutes.

This misclassification produces a characteristic set of effects. Procedural norms are naturalized and treated as features of the world rather than as revisable rules of coordination, the space of negotiation collapses as alternative interpretations are excluded, and disagreement is reframed as existential rather than practical, transforming what could be resolved through procedural revision into a perceived threat to identity or reality itself.

Understanding this mechanism requires tools beyond standard truth-conditional semantics.

1.3 Relation to Existing Literature

The framework developed here draws on and synthesizes several distinct traditions.

In philosophy of language, the distinction between sense and reference (Frege), between description and naming (Russell), and between rigid designation and use (Kripke) establishes that reference alone cannot account for meaning Frege [1892], Russell [1905], Kripke [1980]. Donnellans distinction between referential and attributive uses of descriptions further destabilizes a purely referential account Donnellan [1966].

Speech act theory and performativity shift attention from truth to action. Austins performatives and Searles institutional facts demonstrate that many utterances do not describe the world but change it Austin [1962], Searle [1969]. However, these accounts stop short of formalizing long-range constraint on future states.

Inferentialist and pragmatist approaches, especially Brandoms account of meaning as normative inferential role, come closer to the present view Brandom [1994]. Yet they lack a fully explicit temporal or event-based semantics.

In legal and institutional theory, Harts distinction between primary and secondary rules reveals how authority emerges from constraint rather than fact Hart [1961]. More recent work in institutional design and governance confirms that symbolic stability depends on enforceable procedural limits, not metaphysical agreement.

Finally, in computer science, operational semantics, game semantics, and event-sourced systems already treat meaning as execution and constraint rather than description. These traditions supply the formal backbone of the present approach.

This paper unifies these strands into a single, explicit framework.

1.4 Methodology

The methodology of this paper is formal, analytic, and comparative.

First, it introduces a precise semantic distinction between reference and constraint, expressed using operator semantics over event histories.

Second, it develops an operational semantics in which meaning is evaluated by its effect on admissible futures rather than by truth conditions.

Third, it lifts this operational account into a categorical framework, allowing structural comparison between symbolic systems via functors and natural transformations.

Throughout, formal development is interleaved with concrete examples drawn from law, organizational systems, and computation. These examples are not illustrative ornaments; they function as empirical anchors for the formal claims.

The approach is intentionally neutral with respect to metaphysical commitments. The framework does not deny reference, belief, or interpretation. It specifies their proper semantic location.

1.5 Contributions and Roadmap

This paper advances several interconnected contributions. It formally distinguishes between operator symbols and referential symbols, and shows how systematic misclassification between these semantic

types gives rise to persistent and intractable conflict. It introduces an event-historical operational semantics in which symbolic meaning is constituted by irreversible constraints on admissible futures rather than by static denotation. It further develops a functorial semantics that enables structural comparison between symbolic systems at an abstract level, independent of their surface representations. In addition, the analysis identifies structural risks inherent in highly expressive operator systems, particularly with respect to domination, lock-in, and loss of negotiability. Finally, the framework is shown to apply across legal, organizational, and computational domains, demonstrating its generality and practical relevance.

The paper proceeds as follows. Section 2 deepens the philosophical foundations of the reference/operator distinction. Section 3 develops a formal account of semantic compression and drift. Section 4 analyzes reification as a category error and proves its consequences. Section 5 develops narrative as a constraint-preserving encoding mechanism. Sections 6 and 7 present the operational and functorial semantics, respectively. Subsequent sections apply the framework, compare it with alternatives, and discuss methodological limits and future directions. The conclusion synthesizes the results and outlines practical and theoretical implications.

2 Names, Reference, and Operators

This section develops the conceptual foundation for the papers central distinction between *referential* expressions and *operator* expressions. The distinction is motivated philosophically, formalized semantically, and illustrated across legal, institutional, and computational domains. The aim is not to deny the existence or utility of reference, but to establish that many conflicts arise when expressions whose primary function is to constrain action are mistakenly treated as denoting objects.

2.1 Classical Accounts of Reference

Modern philosophy of language begins with the problem of reference. Frege's distinction between sense (*Sinn*) and reference (*Bedeutung*) establishes that linguistic meaning cannot be reduced to denotation alone Frege [1892]. A name may refer to an object while differing in cognitive content, indicating that reference is only one dimension of meaning.

Russell's theory of descriptions further complicates the picture by showing that many apparently referential expressions function quantificationally rather than denotationally Russell [1905]. Kripke's account of rigid designation restores a strong notion of reference for proper names, but explicitly limits its scope Kripke [1980]. Even on Kripke's account, not all grammatically name-like expressions are rigid designators.

Donnellan's distinction between referential and attributive uses of descriptions demonstrates that reference is often a matter of use rather than form Donnellan [1966]. The same expression may function referentially in one context and non-referentially in another.

These developments collectively undermine the assumption that naming implies objecthood.

2.2 From Reference to Operation

While the classical literature focuses on reference, a parallel tradition shifts attention to what expressions *do*. Austins theory of speech acts demonstrates that many utterances perform actions rather than describe states of affairs Austin [1962]. Searle extends this insight to institutional facts, showing how rules create roles, permissions, and obligations Searle [1969].

However, speech act theory typically treats performative force as localized to utterance events. What remains underdeveloped is an account of how symbolic expressions impose long-range constraints on future action.

To address this gap, we introduce the concept of a linguistic operator.

Definition 1 (Linguistic Operator). *A linguistic operator is a symbolic expression whose primary semantic function is to constrain the space of admissible actions, interpretations, or identities across a structured practice, rather than to denote an object or state of affairs.*

Operators may be realized linguistically, procedurally, or formally. Their meaning lies in their effect on permissible continuation, not in correspondence to an external entity.

2.3 Examples Across Domains

The operator/reference distinction appears across many domains.

Legal Domain. Terms such as *jurisdiction*, *authority*, and *standing* do not denote physical objects. They function as operators that delimit who may act, in what capacity, and under which conditions. Treating such terms as referential properties invites absolutist conflict; treating them as procedural constraints enables negotiation and reform Hart [1961].

Institutional Roles. Expressions like *citizenship* or *membership* regulate access to rights and obligations. Their semantic force derives from rule systems, not from intrinsic properties of individuals. Reifying these roles as natural kinds produces exclusionary pathologies.

Computational Systems. In programming languages, constructs such as `lock`, `scope`, or `permission` are operators over execution paths. They constrain possible program states without denoting runtime objects. Misunderstanding these constructs as values rather than control structures leads to design errors.

These examples illustrate that operator semantics is neither marginal nor exotic. It is pervasive.

2.4 A Taxonomy of Operator Types

Operators may be classified according to the type of constraint they impose on admissible continuation.

Definition 2 (Operator Taxonomy). *Let $\mathcal{H}$ denote a space of admissible event histories. Operators may be distinguished by the manner in which they constrain this space. Some operators function by excluding subsets of $\mathcal{H}$, thereby prohibiting certain future trajectories. Others operate by authorizing*

actions that would otherwise be excluded under existing constraints. A further class imposes precedence relations, introducing ordering constraints among events without eliminating them outright. Identification operators act by equating or merging histories, roles, or trajectories that were previously distinct, while forgetting operators erase internal distinctions among histories while preserving admissibility at a coarser level of resolution.

Each class of operator constrains continuation in a structurally distinct way. Crucially, none of these operators function by denoting objects; their semantic force lies entirely in how they reshape the space of admissible futures.

2.5 Criteria for Classification

To determine whether a symbol functions primarily as an operator or as a referential expression, the framework proposes a set of diagnostic criteria. A symbol exhibits operator-like behavior when modifications to its use alter what may occur next without changing the factual description of the present state, when its interpretation depends essentially on institutional or procedural context, and when disputes concerning its application can be resolved through rule revision rather than by appeal to empirical discovery. A further indicator is temporal reach: symbols that primarily constrain future states of affairs rather than describe present ones are characteristic of operator semantics.

Symbols satisfying most of these criteria should therefore be analyzed as operators rather than as referents.

2.6 Borderline and Hybrid Cases

Some expressions exhibit mixed behavior. For example, *property* may refer to an asset while simultaneously functioning as an operator regulating use and transfer. Hybrid cases require careful semantic typing to avoid category error.

Remark 1. *Hybrid symbols are especially prone to reification errors because their referential aspect masks their operator function.*

2.7 Historical Development of the Distinction

The gradual recognition of operator semantics parallels broader shifts in analytic philosophy away from static correspondence models toward use- and practice-based accounts of meaning. Wittgenstein's later work emphasizes rule-following and language games Wittgenstein [1953], while Brandom formalizes meaning in terms of normative inferential roles Brandom [1994].

The present framework extends these insights by introducing an explicit temporal and structural account of constraint, enabling precise analysis of persistence, drift, and conflict.

2.8 Section Summary

This section has argued that many grammatically name-like expressions function primarily as operators rather than referents. Misclassifying such expressions produces structural conflict by

converting procedural constraints into alleged objects of dispute. Establishing the operator/reference distinction is therefore a prerequisite for the formal analysis that follows.

3 Semantic Compression and Drift

This section develops a rigorous account of semantic compression and drift. The central claim is that symbolic systems preserve coordination by compressing high-dimensional normative and procedural constraints into low-dimensional representations. While such compression is necessary for scalability and stability, it introduces predictable failure modes. Over time, compressed symbols drift away from their original operator structure, increasing the risk of reification and conflict.

The analysis proceeds in four stages. First, semantic compression is formalized using information-theoretic tools. Second, historical case studies illustrate long-term drift in institutional symbols. Third, a typology of drift mechanisms is developed. Finally, semantic drift is modeled as a dynamical process over constraint spaces.

3.1 Compression as an Information-Theoretic Necessity

Let a symbolic system be understood as a mapping from complex constraint structures to manageable representations. Coordination across agents requires that these representations be communicable, memorable, and repeatable.

Formally, let X denote a space of high-dimensional constraint states, and let Y denote a space of symbolic representations. A semantic compression is a mapping

$$C : X \rightarrow Y$$

that reduces descriptive complexity while preserving sufficient information for coordination.

Using Shannon entropy, the compression ratio may be characterized as

$$\kappa = \frac{H(Y)}{H(X)},$$

where $H(\cdot)$ denotes entropy. For long-lived symbolic systems, $\kappa \ll 1$ is not optional but necessary.

However, compression is lossy. Information about the original constraint topology is discarded. The meaning of a symbol is therefore underdetermined by its form alone.

Definition 3 (Semantic Compression). *Semantic compression is the lossy encoding of a high-dimensional constraint structure into a low-dimensional symbolic representation that preserves coordination-relevant information while discarding contextual detail.*

This definition applies equally to legal terms, organizational roles, technical abstractions, and computational interfaces.

3.2 Decompression and Interpretation

Symbols are not used directly. They are interpreted, or decompressed, back into constraints on action.

Let

$$D_t : Y \rightarrow X$$

be a decompression function at time t . Crucially, D_t is not fixed. It depends on background practices, incentives, and institutional context.

Meaning at time t is therefore given by the composite

$$D_t \circ C.$$

Semantic drift occurs when D_t changes while C remains fixed. Because the compressed symbol does not encode its original context, reinterpretation is unavoidable.

3.3 Historical Case Studies of Drift

Constitutional Terminology. Terms such as *due process* or *equal protection* originated as procedural constraints on governance. Over centuries, they have accumulated layers of interpretation, judicial precedent, and political pressure. Disputes over their meaning are rarely resolved by discovering facts; they are resolved by reconfiguring operator structure through reinterpretation and amendment.

Scientific Concepts. Scientific terms such as *gene*, *information*, or *fitness* have undergone substantial drift as empirical knowledge and theoretical frameworks evolved. Early compressed meanings proved insufficient, leading to contested reinterpretations. Treating these terms as fixed referents rather than evolving operators has repeatedly generated confusion.

Organizational Roles. Roles such as *manager* or *fiduciary* encode complex responsibility structures. As organizations scale and technologies change, these compressed symbols drift away from their original constraint sets, producing governance failures unless explicitly re-articulated.

These cases demonstrate that drift is not accidental; it is structural.

3.4 Mechanisms of Semantic Drift

Semantic drift arises through multiple interacting mechanisms that alter how compressed symbolic structures are interpreted over time.

Definition 4 (Drift Mechanisms). *Semantic drift refers to systematic changes in the interpretation of symbolic operators induced by shifts in the conditions of their use. Common mechanisms include the loss of background practices that originally anchored interpretation, reinterpretation driven by misaligned incentives or strategic advantage, the imposition of preferred decompressions by agents or institutions with asymmetric power, changes in feasible action brought about by new technological*

affordances, and the gradual accumulation of institutional modifications that layer partial adjustments rather than re-articulating underlying constraints.

Each mechanism alters D_t without modifying the symbol itself, increasing semantic divergence across agents.

3.5 Drift as a Dynamical System

Let D_t evolve according to a dynamical process

$$D_{t+1} = F(D_t, I_t, P_t),$$

where I_t represents incentive structures and P_t represents power distributions.

The resulting system may exhibit a range of dynamical behaviors over time, including convergence toward a stable interpretation, oscillation corresponding to persistently contested meaning, divergence in which interpretations fragment across subcommunities, and lock-in characterized by path-dependent rigidity. In this dynamical perspective, reification corresponds to a particular attractor in which an operator is treated as a fixed referent, thereby freezing the evolution of the admissible future space D_t and eliminating the possibility of further negotiation.

Remark 2. *Reification is not semantic stability but semantic arrest.*

3.6 Compression–Drift Tradeoff

There exists a fundamental tradeoff between compression efficiency and drift risk. Highly compressed symbols scale well but drift rapidly; richly articulated symbols resist drift but scale poorly.

This tradeoff can be expressed schematically as:

$$\text{Scalability} \uparrow \iff \text{Drift Risk} \uparrow$$

Managing symbolic systems therefore requires periodic re-articulation of operator structure.

3.7 Section Summary

Semantic compression is necessary for coordination but introduces unavoidable drift. Drift becomes pathological when compressed operator symbols are treated as referential absolutes, producing reification and conflict. Understanding drift as a dynamical process prepares the ground for the formal analysis of category error in the next section.

4 Reification and Category Error

This section analyzes reification as a precise semantic failure: a category error in which linguistic operators are misclassified as referential expressions. The analysis proceeds by formal definition, proof of consequences, examination of boundary cases, and articulation of strategies for reversing reification once it has occurred.

The central result is that reification transforms negotiable constraint systems into absolutist symbolic regimes by collapsing procedural disagreement into existential opposition.

4.1 Formal Definition of Reification

Let $\mathcal{L}$ be a language whose expressions are typed into distinct semantic categories. We distinguish in particular between two primary types. Expressions of type $\mathcal{R}$ are referential, and their semantic evaluation proceeds by correspondence to objects, properties, or states of affairs. Expressions of type $\mathcal{O}$ are operators, and their semantic contribution consists in constraining the space of admissible future continuations of an event history rather than in denoting entities.

Definition 5 (Semantic Typing). *A semantic typing is a function*

$$\tau : \mathcal{L} \rightarrow \{\mathcal{R}, \mathcal{O}\}$$

assigning each expression its dominant semantic role.

Definition 6 (Reification Error). *A reification error occurs when an expression σ such that $\tau(\sigma) = \mathcal{O}$ is treated, interpreted, or adjudicated as if $\tau(\sigma) = \mathcal{R}$.*

Reification is thus not a metaphysical mistake but a semantic one.

4.2 Formal Consequences of Reification

We now prove the central proposition introduced earlier.

Proposition 1 (Reification Produces Absolutism). *If a symbolic system treats its highest-order operators as referential objects, then disagreement within that system tends toward absolutism rather than negotiation.*

Proof. Let σ be an operator constraining admissible histories $\sigma : \mathcal{H} \rightarrow \mathcal{P}(\mathcal{H})$. As an operator, disagreement over σ concerns which constraints to impose and is resolvable by modification, composition, or reinterpretation.

If σ is retyped as referential, disagreement over σ becomes a dispute over the existence, identity, or nature of an object. Referential disagreement admits no procedural resolution; it can only be settled by denial or elimination of opposing claims.

Thus, the space of negotiable futures collapses. Disagreement becomes existential rather than procedural. □ □

4.3 Corollaries

[Collapse of Negotiation Space] Reification strictly reduces the set of admissible resolutions available to a symbolic system.

[Path Dependence] Once reification occurs, subsequent reinterpretations accumulate asymmetrically, producing lock-in effects.

[Conflict Amplification] Reified operators amplify conflict by converting rule disagreement into identity threat.

Each corollary follows directly from the elimination of procedural degrees of freedom.

4.4 Boundary Conditions and Counterexamples

Not all symbols are susceptible to reification in the same way.

Pure Referents. Expressions denoting physical objects or well-defined quantities are correctly typed as referential. Treating them as operators produces confusion but not absolutism.

Explicitly Typed Operators. Systems that explicitly distinguish rules from facts such as typed programming languages or well-designed legal codes exhibit strong resistance to reification.

Hybrid Symbols. Symbols combining referential and operator aspects require careful handling. Failure to disambiguate their roles produces instability rather than immediate absolutism.

These cases delimit the scope of the theory.

4.5 Cognitive and Social Mechanisms of Reification

Reification is reinforced by well-documented cognitive and social tendencies.

Cognitive Load Reduction. Treating operators as objects simplifies reasoning by reducing temporal and procedural complexity.

Essentialist Bias. Humans exhibit a bias toward treating abstract roles as natural kinds, even when they are rule-defined.

Institutional Incentives. Power asymmetries reward actors who present procedural constraints as immutable facts.

These mechanisms explain why reification is common even when formally unjustified.

4.6 De-Reification: Recovering Operator Structure

Despite its stabilizing appearance, reification is reversible.

Definition 7 (De-Reification). *De-reification is the semantic operation of restoring an expressions operator typing by re-articulating the constraints it imposes on admissible futures.*

Effective strategies for de-reification involve making governing rules explicit, reframing disputes in procedural rather than referential terms, and decomposing constraints temporally so that their effects on admissible futures can be examined over time. In addition, reconstructing the original operator intent through narrative analysis can restore the regulative function of symbols that have become reified, thereby reopening interpretive and negotiable space.

Crucially, de-reification does not require consensus on interpretation only agreement that the symbol functions procedurally.

4.7 Formal Characterization of De-Reification

Let σ be reified. De-reification corresponds to replacing a referential judgment

$$\sigma = x$$

with an operator judgment

$$\sigma(h) = \{h' \mid h' \text{ satisfies constraint } C\}.$$

This restores negotiability by reopening the space of admissible histories.

Remark 3. *De-reification reintroduces time into meaning. Absolutism is timeless; procedure is temporal.*

4.8 Section Summary

Reification is a semantic category error with predictable structural consequences. By mistaking operators for referents, symbolic systems collapse negotiation into absolutism. Recognizing and reversing this error is a necessary precondition for durable coordination. The next section examines how narrative serves as a constraint-preserving alternative to reification.

5 Narrative as Constraint-Preserving Encoding

The preceding sections analyzed the failure modes that arise when symbolic operators are reified as referents. This section develops the complementary thesis: narrative functions as a robust mechanism for preserving operator structure without collapsing it into absolutist form. Narrative encodes constraints indirectly, distributing semantic force across time, agents, and consequence rather than locating it in isolated propositions.

This section formalizes narrative as a constraint-preserving encoding over event histories, compares it to alternative preservation mechanisms, and explains its central role in maintaining negotiability in long-lived symbolic systems.

5.1 Narrative Versus Propositional Encoding

Propositional encoding represents meaning by asserting facts. Narrative encoding represents meaning by exhibiting sequences of events subject to constraints.

Let $\mathcal{H}$ denote a space of event histories. A propositional statement partitions $\mathcal{H}$ into histories where the proposition holds and histories where it does not. By contrast, a narrative selects a structured subset of $\mathcal{H}$ whose admissible continuations implicitly encode norms.

Definition 8 (Narrative Encoding). *A narrative encoding of a constraint C is a structured projection*

$$N \subseteq \mathcal{H}$$

such that admissible interpretations of N enforce C without explicitly asserting it as a proposition.

Narratives thus preserve constraint topology while avoiding explicit operator invocation.

5.2 Temporal Structure and Constraint Distribution

Narratives distribute meaning across temporal structure. No single event or utterance carries the full semantic load. Instead, constraints emerge from patterns of action and consequence.

This distribution has two important effects. First, it resists reification by preventing meaning from being localized in any single symbol, assertion, or moment. Second, it permits multiple decompressions that remain consistent with the same underlying operator structure, allowing interpretive flexibility without sacrificing semantic coherence.

Formally, let $N = \langle e_1, \dots, e_n \rangle$ be a narrative sequence. The constraint enforced by N is not attributable to any e_i alone but to the relations among them.

Remark 4. *Narrative meaning is irreducibly temporal. Attempts to extract a timeless core invite reification.*

5.3 Narrative and Temporal Logic

Narrative constraints can be formalized using temporal logic. Let $\Box$ denote necessity over futures and $\Diamond$ possibility.

Rather than asserting $\Box P$ directly, a narrative may demonstrate that any history in which P fails leads to undesirable or excluded continuations. The constraint is encoded by exhibited consequence rather than asserted necessity.

This aligns narrative semantics with model-checking rather than truth evaluation.

5.4 Comparison with Other Preservation Mechanisms

Narrative is not the only method for preserving operator structure, but it has distinct advantages.

Ritualized Procedure. Procedures encode constraints through repetition but are brittle under context change.

Case Law and Precedent. Precedent preserves constraint through exemplars, closely approximating narrative structure. Its success depends on analogical reasoning.

Formal Specification. Formal systems preserve constraints precisely but scale poorly across human interpretive communities.

Narrative occupies a unique position: high interpretive flexibility with constraint stability.

5.5 Examples Across Domains

Legal Reasoning. Judicial opinions often rely on narrative reconstruction of facts to justify procedural outcomes. The binding force lies not in explicit rule citation but in the exhibited pattern of reasoning.

Organizational Memory. Organizations transmit norms through stories of success and failure. These stories encode constraints on acceptable behavior without issuing formal directives.

Simulation and Training. Simulations encode constraints by allowing agents to explore outcome spaces. Narrative scenarios function as executable operator semantics.

5.6 Narrative and Reinterpretation

Because narrative avoids explicit operator invocation, it supports reinterpretive plasticity. Different agents may extract different surface meanings while preserving deep constraint structure.

Let D_i and D_j be two decompression functions applied to the same narrative N . If both yield constraint-equivalent interpretations, coordination remains possible despite disagreement.

Definition 9 (Constraint-Equivalent Interpretations). *Two interpretations are constraint-equivalent if they induce the same admissible future space.*

This explains how narrative sustains coordination across heterogeneous agents.

5.7 Computational Perspectives

From a computational standpoint, narrative corresponds to execution traces in which constraints are inferred from the distinction between reachable and unreachable states rather than asserted propositionally. On this view, narrative semantics aligns naturally with formalisms such as process algebras, game semantics, reinforcement learning environments, and scenario-based planning, each of which characterizes meaning in terms of permitted trajectories through a state space. In all such cases, semantic content is not given by isolated assertions but emerges from the structure of possible and impossible paths.

5.8 Narrative as Anti-Reification Mechanism

Narrative resists reification by distributing meaning across temporal structure, by embedding constraints in patterns of consequence rather than in isolated assertions, and by preventing individual symbols from being detached from the contexts that give them regulative force. Attempts to extract a single authoritative proposition from a narrative typically destroy its semantic function, replacing constraint-guided interpretation with a brittle and misleading referential reading.

Remark 5. *Narrative fails only when it is forced to answer referential questions it was not designed to answer.*

5.9 Section Summary

Narrative functions as a constraint-preserving encoding that avoids the category errors associated with reification. By distributing semantic force across time and consequence, it stabilizes operator structure while maintaining negotiability. This makes narrative indispensable to long-lived symbolic systems. The next sections formalize these insights in operational and categorical terms.

6 Operational Semantics on Event Histories

This section provides a complete operational semantics for symbolic meaning understood as constraint on admissible futures. Rather than assigning truth conditions to expressions, the semantics specifies how symbolic operators act on event histories to restrict or transform the space of possible continuations.

Operational semantics is chosen for two reasons. First, it makes temporal irreversibility explicit. Second, it aligns semantic evaluation with execution, rendering meaning directly actionable.

6.1 Event Histories

Let $\mathcal{H}$ be a set of finite, prefix-closed sequences of events. An event history $h \in \mathcal{H}$ is written as

$$h = \langle e_1, e_2, \dots, e_n \rangle,$$

where each e_i represents an irreversible update.

Definition 10 (Prefix Order). *For histories $h, h' \in \mathcal{H}$, we write $h \preceq h'$ if h is a prefix of h' .*

The prefix order induces a partial order on $\mathcal{H}$ representing irreversible extension.

Definition 11 (Future Space). *For a history h , the future space is*

$$\text{Fut}(h) = \{h' \in \mathcal{H} \mid h \preceq h'\}.$$

Once an event occurs, histories incompatible with it are permanently excluded from $\text{Fut}(h)$.

6.2 Operators as Transition Constraints

An operator acts by eliminating or restructuring futures rather than by adding content to history.

Definition 12 (Operator). *An operator is a function*

$$\sigma : \mathcal{H} \rightarrow \mathcal{P}(\mathcal{H})$$

such that $\sigma(h) \subseteq \text{Fut}(h)$ for all $h \in \mathcal{H}$.

Operators are monotone: they cannot introduce futures incompatible with the current history. [Monotonicity] For any operator σ and histories $h \preceq h'$,

$$\sigma(h') \subseteq \sigma(h).$$

Proof. Any future admissible after h' must also be admissible after h , since h' introduces additional constraints. □ □

6.3 Configurations and Rewrite Rules

Semantic evaluation proceeds by rewriting configurations.

Definition 13 (Configuration). *A configuration is a pair (h, Σ) where $h \in \mathcal{H}$ and Σ is a multiset of active operators.*

A rewrite relation

$$(h, \Sigma) \longrightarrow (h', \Sigma')$$

specifies a single semantic step.

6.4 Core Rewrite Schemes

We define representative rewrite schemes corresponding to common operator types.

POP (Exclusion).

$$\frac{h' \in \text{Fut}(h) \quad h' \notin \sigma(h)}{(h, \{\sigma\}) \longrightarrow (h, \emptyset)}$$

POP eliminates incompatible futures.

BIND (Precedence).

$$(h, \{\text{BIND}(e_i \prec e_j)\}) \longrightarrow (h, \Sigma')$$

where Σ' enforces the ordering constraint on all extensions of h .

MERGE (Identification).

$$(h_1, h_2, \{\text{MERGE}\}) \longrightarrow (h, \Sigma)$$

where h is the least upper bound of h_1 and h_2 if one exists.

COLLAPSE (Forgetting).

$$(h, \{\text{COLLAPSE}\}) \longrightarrow (h', \Sigma)$$

where h' is a quotient of h under an equivalence relation preserving admissibility.

Each rule eliminates or restructures futures without adding new ones.

6.5 Soundness

[Soundness] If $(h, \Sigma) \longrightarrow (h', \Sigma')$, then

$$\text{Fut}(h') \subseteq \text{Fut}(h).$$

Proof. Each rewrite rule either preserves h or replaces it with a quotient or merge consistent with h . No rule introduces histories incompatible with prior events. Therefore admissible futures are monotonically reduced. $\square$ $\square$

Soundness ensures that semantic evaluation corresponds to constraint strengthening.

6.6 Completeness

Completeness ensures that all constraint reductions can be realized operationally.

[Completeness] For any operator σ and history h , if $h' \in \sigma(h)$, then there exists a finite rewrite sequence

$$(h, \{\sigma\}) \longrightarrow^* (h', \Sigma')$$

realizing the constraint imposed by σ .

Proof. By construction, operators specify admissible futures. Each elimination or restructuring step corresponds to a rewrite rule. Finite composition suffices to exclude all incompatible histories. $\square$ $\square$

Completeness guarantees that operator semantics is executable.

6.7 Worked Examples

Consider a history $h = \langle e_1 \rangle$ and an operator enforcing that event e_2 may occur only after e_3 .

Applying BIND yields:

$$(h, \{\text{BIND}(e_3 \prec e_2)\}) \longrightarrow (h, \Sigma')$$

where Σ' excludes all futures in which e_2 precedes e_3 .

The resulting future space contains only histories consistent with the constraint. No propositional assertion is required.

6.8 Comparison with Other Semantic Frameworks

Denotational Semantics. Denotational approaches assign mathematical objects to expressions. While powerful, they obscure temporal irreversibility and negotiation dynamics.

Modal Semantics. Modal logic captures necessity and possibility but treats modality statically. Operator semantics embeds modality in execution.

Game Semantics. Game semantics models interaction explicitly. Operator semantics may be viewed as a restriction of game semantics to admissible trajectories.

Operational semantics uniquely combines temporal structure with constraint evaluation.

6.9 Algebraic Properties of Operators

Operators compose naturally.

Definition 14 (Operator Composition). *Given operators σ_1, σ_2 , define*

$$(\sigma_2 \circ \sigma_1)(h) = \sigma_2(\sigma_1(h)).$$

Composition is associative but not necessarily commutative. Identity operators act as neutral elements.

These properties prepare the ground for categorical abstraction.

6.10 Section Summary

This section established a complete operational semantics for meaning as constraint on event histories. Soundness and completeness ensure that semantic evaluation is both faithful and executable. The next section lifts this framework to a categorical level, enabling structural comparison between symbolic systems.

7 From Operator Semantics to Event–Historical Structure

This appendix provides a precise correspondence between the operator-based semantic framework developed above and an event-historical semantics in which meaning, identity, and obligation are constituted by irreversible transitions.

7.1 Event–Historical Substrate

Let $\mathcal{H}$ denote a partially ordered set of events equipped with a precedence relation $\prec$. An event history $h \in \mathcal{H}$ is a finite or countably infinite prefix-closed sequence of events.

Definition 15 (Event). *An event is an irreversible update that reduces the space of admissible future histories.*

Irreversibility is essential: an event does not merely describe change but *settles* what can no longer occur.

7.2 Operators as Constraints on Futures

Recall that an operator was defined as a function constraining admissible action. We now refine this definition in event-historical terms.

Definition 16 (Event-Historical Operator). *An operator σ is a function*

$$\sigma : \mathcal{H} \rightarrow \mathcal{P}(\mathcal{H})$$

such that for any history h , $\sigma(h)$ is the set of admissible future extensions of h consistent with σ .

Operators do not add content to history; they restrict continuation. Their semantic force is entirely future-directed.

This formalizes the intuition that sacred or legal symbols regulate conduct rather than denote entities.

7.3 Spherepop Correspondence

In Spherepop-style calculi, existence and identity arise from explicit operations rather than axioms. We may map core Spherepop primitives to the present framework as follows.

Spherepop Operation	Semantic Interpretation
POP	Irreversible elimination of future branches
MERGE	Identification of histories under shared constraints
BIND	Introduction of precedence constraints on events
COLLAPSE	Erasure of historical differentiation

Each operation corresponds to a transformation on $\mathcal{H}$ that alters the space of admissible continuations.

Proposition 2. *Spherepop operations are higher-order operators acting on event histories rather than on values or propositions.*

Proof. Each operation modifies $\mathcal{P}(\mathcal{H})$ by restricting, identifying, or releasing future extensions. No operation asserts a truth condition; all impose structural constraints. $\square$

7.4 Identity as Stabilized History

Under this mapping, identity is not primitive.

Definition 17 (Historical Identity). *An identity is an equivalence class of event histories stabilized under a set of operators.*

Names, titles, and symbolic markers do not refer to entities but to stabilized regions of $\mathcal{H}$. Reification occurs when such equivalence classes are mistaken for timeless objects.

This explains why identity-based conflicts intensify when symbolic operators are treated as ontological predicates.

8 A Functorial Semantics of Constrained Futures

The previous section defined meaning operationally as constraint on admissible event histories. This section abstracts that account categorically. The resulting functorial semantics makes explicit which structural features of symbolic systems are invariant under reinterpretation and which are not.

Category theory is employed not as an added layer of abstraction, but as a means of isolating semantic structure from representational accident. Two symbolic systems are semantically equivalent, on this view, if their constraint structures are categorically isomorphic, regardless of surface vocabulary.

8.1 The Category of Event Histories

Define a category **Hist** whose objects are event histories $h \in \mathcal{H}$ and whose morphisms encode irreversible extension. For any two histories h and h' , there exists a unique morphism $f : h \rightarrow h'$ if and only if $h \preceq h'$, that is, if h is a prefix of h' .

Proposition 3. *The category **Hist** is thin, and hence a preorder, encoding the irreversible extension of event histories.*

Proof. By definition, there is at most one morphism between any two objects. Reflexivity and transitivity follow from prefix inclusion. $\square$ $\square$

The categorical structure directly mirrors temporal irreversibility.

8.2 The Category of Constraint Spaces

Let **Constr** be the category defined by:

- Objects are subsets $C \subseteq \mathcal{H}$ closed under extension (i.e. if $h \in C$ and $h \preceq h'$, then $h' \in C$).
- Morphisms are monotone functions preserving inclusion.

Proposition 4. ***Constr** is a complete lattice category.*

Proof. Arbitrary intersections and unions of extension-closed sets are extension-closed, yielding limits and colimits. $\square$ $\square$

Constr captures admissibility structure independently of presentation.

8.3 The Meaning Functor

We now define the central semantic construction.

Definition 18 (Meaning Functor). *Define a functor*

$$\mathcal{M} : \mathbf{Hist} \rightarrow \mathbf{Constr}$$

by

$$\mathcal{M}(h) = \text{Fut}(h),$$

and for $f : h \rightarrow h'$,

$$\mathcal{M}(f)(C) = C \cap \text{Fut}(h').$$

[Functoriality] $\mathcal{M}$ is a well-defined functor.

Proof. Identity morphisms map to identity inclusions. Composition of prefix morphisms corresponds to successive restriction of admissible futures. Associativity follows from set intersection. $\square$ $\square$

The meaning functor assigns to each history its space of admissible continuations.

8.4 Operators as Endofunctors

Each operator induces an endofunctor on **Constr**.

Definition 19 (Operator Endofunctor). *Given an operator σ , define an endofunctor*

$$F_\sigma : \mathbf{Constr} \rightarrow \mathbf{Constr}$$

by

$$F_\sigma(C) = C \cap \sigma(h),$$

for any history h generating C .

Proposition 5. *Operator composition corresponds to functor composition.*

Proof. Intersection of admissible futures is associative. Identity operators act as identity functors.

□

□

Thus, symbolic systems correspond to monoids of endofunctors.

8.5 Natural Transformations as Reinterpretation

Reinterpretation of symbols corresponds categorically to natural transformation.

Definition 20 (Semantic Reinterpretation). *A reinterpretation of an operator σ as σ' is a natural transformation*

$$\eta : F_\sigma \Rightarrow F_{\sigma'}.$$

Naturality ensures that reinterpretation commutes with history extension.

Remark 6. *Semantic drift corresponds to loss of naturality: reinterpretation behaves differently across contexts.*

This formalizes the intuition that drift is not mere change, but structural misalignment.

8.6 Limits, Colimits, and Identity

MERGE operations correspond to colimits in **Constr**, while COLLAPSE corresponds to coequalizers.

Proposition 6. *Stable identities arise as fixed points of operator endofunctor composition.*

Proof. An identity persists exactly when repeated application of constraints leaves the admissible future space invariant. □ □

Identity is thus an emergent categorical invariant, not a primitive referent.

8.7 Semantic Equivalence

Two symbolic systems are semantically equivalent if their meaning functors are naturally isomorphic.

Definition 21 (Semantic Equivalence). *Two systems S_1 and S_2 are semantically equivalent if there exists a natural isomorphism*

$$\mathcal{M}_1 \cong \mathcal{M}_2.$$

This criterion ignores vocabulary and focuses solely on constraint structure.

Remark 7. *Many apparent metaphysical disagreements reduce to failures of recognizing semantic equivalence.*

8.8 Adjunctions and Relaxation

Relaxation of constraints corresponds to left adjoints; enforcement corresponds to right adjoints.

Definition 22 (Constraint Adjunction). *An adjunction $L \dashv R$ between endofunctors represents reversible negotiation between permissiveness and restriction.*

Adjunctions formalize the possibility of controlled loosening without collapse.

8.9 Section Summary

This section lifted operational semantics into a categorical framework. Meaning is a functor from histories to admissible futures; symbols are endofunctors; and reinterpretation is natural transformation. This abstraction isolates semantic invariants and clarifies when disagreement is substantive and when it is merely presentational.

The next section applies the framework to concrete domains and extended case studies.

9 Applications and Case Studies

The preceding sections developed a formal theory in which meaning is understood as constraint on admissible futures rather than reference to objects. This section demonstrates the explanatory and diagnostic power of that framework across multiple domains. In each case, persistent conflict or instability is shown to arise from reification of operator symbols, and resolution is shown to depend on restoring operator-level interpretation.

The aim is not to offer policy prescriptions, but to show how semantic structure conditions the space of possible resolutions.

9.1 Legal Systems

Legal systems provide a paradigmatic case of operator semantics at scale. Legal language is replete with expressions that regulate conduct without denoting objects.

Constitutional Interpretation. Constitutional terms such as *authority*, *due process*, or *jurisdiction* function as operators that constrain permissible actions of institutions. Disputes framed as disagreements about the true meaning of such terms often persist because the terms are treated as referential absolutes rather than adjustable constraints.

Under an operator semantics, constitutional interpretation becomes a problem of constraint preservation under changing conditions. Amendments, precedent, and doctrinal tests function as reinterpretations rather than discoveries of fixed reference.

Contract Law. Contracts encode future constraints on action. Treating contractual language as descriptive rather than performative leads to brittle enforcement and adversarial interpretation. Operator semantics clarifies why doctrines such as good faith and reasonableness are indispensable: they reintroduce negotiability at the operator level.

9.2 Organizational Governance

Organizations coordinate action through role definitions, procedures, and protocols. These are operator-rich systems.

Role Reification. Roles such as *manager*, *executive*, or *stakeholder* regulate decision authority. When these roles are reified as intrinsic personal properties rather than procedural positions, organizations experience rigidity, blame cycles, and power struggles.

Operator semantics explains why effective organizations continuously revise role definitions: they are re-articulating constraints to maintain adaptability.

Protocol Design. Organizational protocols function as executable narratives. Case escalation procedures, incident response playbooks, and governance charters encode constraints via conditional sequences rather than propositions.

Failures often arise when protocols are treated as literal instructions rather than as constraint-generating structures adaptable to context.

9.3 Computational Systems

Computational systems provide a domain where operator semantics is already implicitly accepted.

Programming Language Semantics. Control structures such as scopes, locks, and permissions constrain execution paths. Treating these constructs as values rather than operators leads to security and concurrency failures.

Operational semantics makes explicit why static typing and access control are forms of semantic constraint enforcement.

Distributed Systems. Consensus protocols do not describe truth; they restrict admissible histories of system state. Operator semantics clarifies why such systems fail under network partition: admissible future spaces fragment.

AI and Multi-Agent Systems. Reward functions and policies are operators over agent trajectories. Reifying objectives as fixed goals rather than adjustable constraints produces brittle and misaligned systems. Operator semantics reframes alignment as constraint negotiation over futures rather than goal specification.

9.4 Conflict Resolution and Mediation

Many entrenched conflicts persist despite agreement on outcomes because parties disagree on symbolic framing.

Operator semantics suggests a diagnostic procedure for addressing persistent symbolic conflict. The procedure begins by identifying symbols that are treated as referential absolutes rather than as regulative devices. It then proceeds by recovering the operator functions those symbols originally served, followed by the explicit re-articulation of the constraints they impose on admissible action. Finally, negotiability is restored through procedural revision, allowing coordination to occur without requiring agreement on underlying metaphysical or interpretive commitments.

This procedure does not require consensus on narratives or identities. It requires only agreement that certain symbols regulate action rather than denote objects.

9.5 Formal Modeling of a Case Study

Let two agents share a history h and disagree over a symbol σ . If σ is treated referentially, admissible futures partition into disjoint sets with no overlap.

Under operator interpretation, both agents agree on a constraint $\sigma : \mathcal{H} \rightarrow \mathcal{P}(\mathcal{H})$, even if they differ on surface interpretation. The intersection

$$\text{Fut}_1(h) \cap \text{Fut}_2(h)$$

remains non-empty, preserving coordination.

This formal distinction explains why reframing disputes procedurally often succeeds where factual argument fails.

9.6 Section Summary

Across legal, organizational, and computational systems, persistent conflict arises when operator symbols are reified. Resolution depends not on discovering correct reference, but on restoring constraint structure. The next section compares this framework with alternative semantic theories and addresses potential objections.

10 Comparison with Alternative Frameworks

This section situates the constraint-first framework relative to established approaches in philosophy of language, semantics, and coordination theory. The goal is not to dismiss alternative frameworks, but to clarify the specific phenomena each explains well, the failure modes they exhibit, and the sense

in which operator semantics provides a complementary and in some cases strictly stronger explanatory account of persistent symbolic conflict.

10.1 Truth-Conditional Semantics

Truth-conditional semantics evaluates meaning by specifying the conditions under which an expression is true or false. This approach has been extraordinarily successful in modeling descriptive language and factual discourse.

However, truth-conditional semantics encounters difficulty in domains where disagreement persists despite agreement on observable facts. In such cases, truth conditions are either trivially satisfied or irrelevant to the dispute.

Operator semantics explains this failure mode: the disputed expressions do not primarily function to describe states of affairs. They regulate admissible actions and identities. Evaluating them by truth conditions misclassifies their semantic role.

Truth-conditional semantics remains appropriate for referential expressions. The present framework argues only that it is incomplete as a general theory of meaning.

10.2 Pragmatist and Inferentialist Accounts

Pragmatist approaches emphasize use, practice, and consequences over reference. Inferentialist theories, particularly those developed by Brandom, understand meaning in terms of normative inferential roles.

These accounts align closely with the present framework. Both reject the primacy of correspondence and emphasize rule-governed practice. However, they typically lack an explicit temporal or event-historical semantics.

Constraint-first semantics extends inferentialist approaches by making the role of irreversibility explicit, by modeling how constraints operate over extended temporal horizons rather than only locally within inferential contexts, and by providing an executable semantics in which meaning can be realized through concrete transitions over event histories. The result is a more precise account of how meaning stabilizes over time and of why semantic misclassification, rather than substantive disagreement, so often gives rise to persistent conflict.

10.3 Speech Act Theory and Performativity

Speech act theory demonstrates that many utterances perform actions rather than describe facts. Performativity captures how language brings about institutional states.

While foundational, speech act theory remains largely local: it focuses on the effect of utterances at the moment of performance. Operator semantics generalizes this insight to persistent, distributed constraint over histories.

In effect, operator semantics treats institutions themselves as extended speech acts whose force unfolds temporally.

10.4 Game-Theoretic and Economic Models

Game-theoretic models analyze coordination in terms of incentives, equilibria, and strategies. These models excel at predicting behavior under fixed rules.

However, they typically assume the rule structure as given. When disputes arise over the rules themselves, game-theoretic analysis must step outside its formal frame.

Operator semantics addresses this meta-level by modeling how rules constrain admissible futures and how those constraints may be renegotiated. It complements game theory by explaining how the game itself is constituted and transformed.

10.5 Realist Objections

A common objection holds that some symbolic disputes concern real facts and cannot be resolved procedurally. The present framework does not deny this. Rather, it provides a criterion for distinguishing factual disagreement from semantic misclassification.

Where disagreement turns on empirical claims, reference remains primary. Where disagreement persists despite empirical convergence, operator semantics offers a diagnostic alternative.

Constraint-first semantics thus narrows the scope of realist claims without rejecting realism outright.

10.6 Summary of Comparative Advantages

The distinctive contributions of the present framework lie in its explicit semantic typing of operators as distinct from referential expressions, its event-historical treatment of meaning as constraint on admissible futures, and its formal modeling of reification and de-reification as structurally identifiable semantic transformations. In addition, it provides categorical criteria for semantic equivalence grounded in preserved constraint structure and offers practical diagnostic tools for analyzing and mitigating persistent symbolic conflict. No existing framework integrates all of these elements within a single, coherent formal account.

10.7 Section Summary

Constraint-first semantics does not replace truth-conditional, pragmatist, or game-theoretic approaches. It explains a class of failures they systematically leave unresolved. The next section addresses methodological considerations, scope, and limitations of the framework.

11 Methodological Considerations

This section addresses methodological questions raised by a constraint-first approach to meaning. It clarifies what neutrality entails in this context, identifies the limits of formal modeling, specifies the scope of applicability, and discusses how the framework can be empirically evaluated. These considerations are essential for situating the theory within broader research programs and for avoiding category errors in its application.

11.1 Neutrality and Non-Commitment

The framework developed in this paper is neutral with respect to metaphysical, ideological, and interpretive commitments. Neutrality here does not mean indifference to outcomes, nor does it imply value skepticism. It means that the theory does not presuppose any particular account of what exists, what is true, or what ultimately matters.

Instead, the framework specifies how symbolic expressions function within rule-governed systems to constrain admissible futures. This functional focus allows the theory to diagnose semantic failures without adjudicating underlying beliefs.

Remark 8. *Neutrality is achieved by semantic typing, not by suspension of judgment.*

By distinguishing operator semantics from referential semantics, the framework avoids implicit metaphysical commitments while preserving analytical precision.

11.2 Limits of Formal Modeling

Formal semantics necessarily abstracts away from aspects of lived practice. While the operational and functorial models capture structural features of constraint, they do not encode affect, persuasion, or power directly.

This limitation is deliberate. The framework is designed to analyze the *conditions of negotiability*, not to model every factor influencing decision-making. Psychological, sociological, and political variables enter the analysis indirectly, through their effect on interpretation functions and power asymmetries.

Formal clarity requires selective abstraction.

11.3 Scope of Applicability

The constraint-first framework applies in contexts where symbols function primarily to regulate action rather than to describe objects, where coordination depends on shared procedural constraints rather than on shared representations, and where disagreement persists even after substantial convergence on empirical facts. In such settings, semantic failure is best diagnosed at the level of constraint interpretation rather than referential disagreement.

These conditions obtain in legal systems, organizational governance, computational protocols, and multi-agent environments. The framework is less useful for domains dominated by empirical discovery or purely descriptive language.

Remark 9. *Applying operator semantics to genuinely referential disputes constitutes a category error.*

The theory therefore includes its own criteria for appropriate use.

11.4 Generalizability and Transfer

Generalizability is achieved structurally rather than substantively. The same formal apparatus applies across domains because it abstracts from domain-specific content and models constraint topology instead.

Transfer between domains requires careful retyping of symbols and explicit identification of admissible histories. Failure to perform this retyping risks importing inappropriate assumptions.

11.5 Empirical Testability

Although the framework is formal, it yields empirically testable predictions.

First, it predicts that conflicts framed in referential terms will exhibit intractability even when procedural consensus exists. Second, it predicts that reframing disputes at the operator level will expand the space of negotiated outcomes.

These predictions can be tested through a combination of experimental studies that manipulate negotiation framing, comparative analyses of institutional reforms across differing semantic regimes, and simulations of multi-agent systems operating under alternative constraint structures. Each of these approaches provides a means of assessing how semantic typing and operator reinterpretation affect the persistence or resolution of conflict.

Empirical validation thus concerns patterns of resolution and persistence, not truth-apt judgments.

11.6 Falsifiability

The framework is falsifiable in several ways. If conflicts diagnosed as reification failures remain intractable after operator-level reframing, the theory is challenged. If referential reframing consistently outperforms procedural reframing in operator-dominated domains, the central claim is undermined.

The theory therefore makes substantive, testable claims about the structure of disagreement.

11.7 Methodological Summary

Constraint-first semantics is a structural, neutral, and formally explicit approach to meaning. Its power lies in diagnosing semantic failure modes and clarifying the conditions under which negotiation is possible. Its limits are those of abstraction itself.

The next section outlines future research directions opened by this framework.

12 Future Directions

The constraint-first framework developed in this paper opens multiple avenues for further research. Some involve deepening the formal apparatus, others involve empirical validation, and still others involve practical deployment in computational and institutional systems. This section outlines these directions and identifies open questions that remain unresolved.

The aim is not to forecast outcomes, but to clarify where the theory's internal logic points next.

12.1 Computational Implementations

One immediate direction is the construction of executable systems that embody operator semantics directly. Event-sourced architectures, rewrite systems, and policy engines already approximate this structure, but often without explicit semantic typing.

Future work may include the development of programming languages that treat operators as first-class semantic entities, the design of runtime systems that explicitly track and manipulate spaces of admissible futures, and the creation of formal verification tools capable of reasoning about the preservation and compatibility of constraints over execution histories. In addition, interpreter designs that make processes of de-reification explicit as program transformations represent a promising direction for rendering semantic structure transparent within computational systems.

Such systems would make semantic structure visible and manipulable, reducing the risk of unintentional reification in software design.

12.2 Multi-Agent Systems and Coordination

The framework naturally extends to multi-agent environments in which coordination depends on shared constraints rather than on shared beliefs or representations. Within such settings, several open questions arise concerning the dynamics of interaction and interpretation. These include how agents negotiate reinterpretations of governing operators, how spaces of admissible futures fragment or merge under conditions of partial agreement, how asymmetries of power shape the distribution and enforcement of constraints, and how stable patterns of coordination emerge through iterative processes of de-reification and reinterpretation.

These questions can be explored through simulation, game-theoretic extension, and experimental economics.

12.3 Empirical Research Programs

Empirical validation of the framework necessarily requires interdisciplinary collaboration, drawing on methods from the social sciences, cognitive science, and computational analysis. Promising lines of inquiry include controlled experimental studies that manipulate negotiation framing, longitudinal analyses of institutional reform processes, corpus-based investigations of symbolic drift in legal or organizational texts, and behavioral studies examining susceptibility to reification bias in interpretive judgment.

Across these approaches, the central empirical variable of interest is not convergence on truth-apt claims, but the expansion or contraction of the space of negotiable futures made available to participants under different semantic framings.

12.4 Formal Extensions

Several formal questions remain open.

Complexity. What is the computational complexity of determining non-empty intersection of admissible future spaces under multiple operators?

Decidability. Under what conditions is constraint compatibility decidable?

Higher-Order Operators. What are the properties of systems in which operators act on other operators?

Topological Structure. Can constraint spaces be given a meaningful topology or metric capturing degrees of negotiability?

Addressing these questions would strengthen the mathematical foundations of the theory.

12.5 Institutional Design

The framework suggests a set of design principles for institutions and protocols concerned with durable coordination. In particular, institutions are more likely to remain adaptable and less prone to absolutist conflict when they explicitly distinguish symbols that function as operators from those that function as referential terms, when they incorporate mechanisms for the periodic de-reification and reinterpretation of their governing symbols, and when they maintain a non-empty space of admissible future actions under revision and change.

The formal articulation and evaluation of these principles constitutes an important direction for future applied research.

12.6 Integration with Other Theories

Constraint-first semantics is compatible with, but distinct from, a range of existing theoretical frameworks. Future work may explore deeper integration with inferentialist approaches to meaning, with models from institutional economics and governance theory that emphasize rule-structured coordination, and with formal epistemology concerned with belief revision and informational dynamics. In addition, the framework invites further engagement with research on AI alignment, where the management of admissible futures and the avoidance of semantic reification are increasingly recognized as central design challenges.

Such integration should proceed cautiously, preserving semantic typing discipline.

12.7 Section Summary

The theory developed here is not complete. It is deliberately foundational. Its value lies in identifying a semantic failure mode and providing tools to analyze and correct it. The directions outlined above represent natural extensions rather than speculative leaps.

The concluding section synthesizes the results and articulates their broader significance.

13 Conclusion

This paper has advanced a simple but far-reaching thesis: in a wide class of symbolic systems, constraint is logically and operationally prior to reference. Many persistent conflicts arise not from deep disagreement about facts, values, or reality, but from a semantic misclassification in which operator symbols are treated as referential absolutes. Once this inversion occurs, negotiable procedural disagreement collapses into existential opposition.

The framework developed here restores category discipline by distinguishing symbols that denote objects from symbols that constrain admissible futures. It does not deny reference, belief, or interpretation. It specifies their proper place.

13.1 Summary of Results

The paper has established the following results:

First, it developed a principled distinction between referential expressions and operator expressions, grounded in the philosophy of language and extended through formal semantics. This distinction explains why many grammatically name-like symbols resist truth-conditional analysis.

Second, it showed that long-lived symbolic systems rely on semantic compression, which is necessary for coordination but introduces predictable drift. Drift becomes pathological when compressed operators are reified as fixed referents.

Third, it provided a formal account of reification as a semantic category error and proved that reification collapses negotiation space by converting procedural constraints into alleged objects of dispute.

Fourth, it demonstrated that narrative functions as a constraint-preserving encoding that resists reification by distributing semantic force across time and consequence.

Fifth, it introduced an operational semantics on event histories in which meaning is executable constraint, and proved soundness and completeness of the resulting rewrite system.

Sixth, it lifted this operational account into a functorial semantics, showing that symbols correspond to endofunctors, reinterpretation to natural transformations, and identity to fixed points of constraint composition.

Seventh, it applied the framework across legal, organizational, computational, and multi-agent domains, demonstrating its diagnostic and explanatory power.

Eighth, it situated the framework relative to existing semantic, pragmatic, and coordination theories, clarifying both its compatibility and its distinct contributions.

13.2 Theoretical Contributions

The principal theoretical contributions of this work can be summarized as follows. The paper introduces a constraint-first ordering of semantic priority that applies across institutional, computational, and interpretive domains. It develops a formal account of reification and de-reification as semantic type errors and corrective reinterpretations, respectively. It advances an event-historical semantics in which meaning is constituted by irreversible constraints on admissible futures rather than by static reference. It provides a categorical criterion for semantic equivalence grounded in preserved constraint structure, and it offers a structural explanation of why certain symbolic conflicts persist despite apparent normative convergence.

Together, these contributions provide a unified account of meaning, stability, and disagreement that neither reduces symbols to reference nor dissolves them into indeterminacy.

13.3 Practical Implications

The practical implications of a constraint-first semantics are substantial.

For legal and institutional systems, the framework clarifies why procedural re-articulation often succeeds where factual argument fails. For organizations, it explains the necessity of revisiting role definitions and protocols. For computational and AI systems, it reframes alignment as the management of admissible futures rather than the specification of fixed objectives.

In each case, stability depends on preserving negotiability at the operator level.

13.4 Ethical Considerations and Misuse

The expressive power of operator semantics entails ethical risk. Systems capable of constraining futures at scale can be used to stabilize cooperation or to eliminate it. The difference lies not in the formalism but in how operator composition is governed.

This paper has therefore emphasized transparency, explicit typing, and preservation of non-empty admissible future space as minimal structural safeguards. These are not moral add-ons; they are conditions for avoiding semantic capture.

13.5 Philosophical Significance

At a broader level, the framework challenges a deeply ingrained assumption in the theory of meaning: that reference is primary and constraint derivative. The analysis here suggests the opposite ordering in many of the domains where meaning matters most for coordination.

Meaning stabilizes action before it stabilizes description. Symbols bind futures before they describe worlds.

Recognizing this priority does not resolve all disagreement. It does something more fundamental: it restores the possibility of resolution by returning conflict to the domain of procedure rather than absolutes.

13.6 Closing Remark

Constraint before reference is not a slogan but a discipline. It is a demand for semantic precision in the places where imprecision carries the highest cost. By recovering the operator structure of symbolic systems, we recover the conditions under which disagreement can remain productive rather than destructive.

That recovery is not optional. It is a precondition for durable coordination in any system complex enough to endure.

A Formal System Specification

This appendix presents the core formal system underlying the constraint-first semantics developed in the main body. The purpose of this appendix is threefold: (1) to make all semantic assumptions explicit, (2) to provide a typed foundation that prevents category errors, and (3) to prepare the ground for operational and categorical proofs in subsequent appendices.

A.1 Semantic Typing Discipline

We begin by fixing the semantic universe and its typing discipline.

A.1.1 Semantic Types

Definition 23 (Semantic Universe). *The semantic universe consists of three disjoint domains:*

$$\mathcal{R} \quad (\text{referential terms}), \quad \mathcal{O} \quad (\text{operators}), \quad \mathcal{H} \quad (\text{event histories}).$$

These domains are semantically disjoint. In particular, operators are not objects, and histories are not propositions.

Remark 10. *Semantic typing is independent of grammatical form. A syntactic noun phrase may denote an element of $\mathcal{O}$ rather than $\mathcal{R}$.*

This separation is the formal mechanism by which reification errors are detected.

A.2 Event Histories

Event histories encode irreversible structure.

Definition 24 (Event). *An event e is an atomic, irreversible update that eliminates at least one possible future continuation.*

Definition 25 (Event History). *An event history h is a finite or countably infinite sequence*

$$h = \langle e_1, e_2, \dots \rangle$$

equipped with a precedence relation that is transitive and antisymmetric, and that satisfies prefix closure. These conditions ensure that the ordering of events is well-defined, irreversible, and compatible with the interpretation of histories as settled pasts extended by admissible futures.

Definition 26 (Prefix Order). *For histories $h, h' \in \mathcal{H}$, define*

$$h \preceq h' \iff h \text{ is a prefix of } h'.$$

Prefix order encodes irreversibility: once an event occurs, all incompatible histories are excluded.

A.3 Admissible Futures

Definition 27 (Future Space). *For a history h , define its future space as*

$$\text{Fut}(h) = \{h' \in \mathcal{H} \mid h \preceq h'\}.$$

[Monotonicity of Futures] If $h \preceq h'$, then

$$\text{Fut}(h') \subseteq \text{Fut}(h).$$

Proof. Any continuation of h' is necessarily a continuation of h , since h is a prefix of h' . No new futures are introduced by extension. $\square$

This lemma formalizes temporal asymmetry.

A.4 Operators as Future Constraints

Definition 28 (Operator). *An operator is a function*

$$\sigma : \mathcal{H} \rightarrow \mathcal{P}(\mathcal{H})$$

such that

$$\sigma(h) \subseteq \text{Fut}(h) \quad \text{for all } h \in \mathcal{H}.$$

Operators do not generate events. They constrain which future histories remain admissible.

[Operator Monotonicity] If $h \preceq h'$, then

$$\sigma(h') \subseteq \sigma(h).$$

Proof. Since $\text{Fut}(h') \subseteq \text{Fut}(h)$ and $\sigma(h') \subseteq \text{Fut}(h')$, it follows that $\sigma(h') \subseteq \text{Fut}(h)$. Operators cannot re-admit excluded futures. $\square$

A.5 Operator Composition

Definition 29 (Operator Composition). *Given operators $\sigma_1, \sigma_2 \in \mathcal{O}$, define their composition by*

$$(\sigma_2 \circ \sigma_1)(h) = \sigma_2(h) \cap \sigma_1(h).$$

Proposition 7 (Closure). *The set $\mathcal{O}$ is closed under composition.*

Proof. Intersection preserves admissibility: both $\sigma_1(h)$ and $\sigma_2(h)$ are subsets of $\text{Fut}(h)$, hence so is their intersection. $\square$

Composition is commutative but not necessarily idempotent.

A.6 Reification as a Type Error

Definition 30 (Reification Error). *A reification error occurs when an expression of semantic type $\mathcal{O}$ is treated as if it were of type $\mathcal{R}$.*

[Semantic Collapse Under Reification] If an operator σ is reified, then its interpretation no longer constrains $\text{Fut}(h)$ but instead introduces truth-conditional evaluation.

Proof. Truth-conditional evaluation presupposes denotation. Operators lack denotation and only define admissibility. Reification therefore discards constraint semantics and replaces it with an inapplicable evaluation regime. $\square$

This formalizes why reification collapses negotiation space: admissibility is replaced by alleged facticity.

A.7 Minimal Consistency Conditions

Definition 31 (Semantic Consistency). *A symbolic system is semantically consistent if and only if for all histories h ,*

$$\bigcap_{\sigma \in \Sigma} \sigma(h) \neq \emptyset.$$

Remark 11. *Consistency here is structural, not logical. It concerns future availability, not contradiction.*

A.8 Appendix Summary

This appendix has established a typed semantic universe in which referential expressions, operators, and event histories are formally distinguished. Within this universe, event histories were characterized as irreversible structures, and operators were defined as constraints on admissible future continuations rather than as denotational entities. Reification was identified as a formal type error arising from the misclassification of operators as referential terms, and semantic consistency was defined in structural terms as the preservation of a non-empty future space.

These definitions provide a sufficient foundation for the operational semantics and accompanying proofs developed in subsequent sections, ensuring that later results rest on a precise and internally coherent formal basis.

B Operational Semantics and Expanded Proofs

This appendix develops the operational semantics implicit in the main text and provides detailed proofs of soundness, completeness, and related properties. Where the main body emphasized conceptual structure, this appendix makes the execution model explicit.

B.1 Configurations and Execution State

B.1.1 Configurations

Definition 32 (Configuration). *A configuration is an ordered pair (h, Σ) , where $h \in \mathcal{H}$ denotes the current event history and $\Sigma \subseteq \mathcal{O}$ is a finite multiset of operators active at that history. The configuration represents a semantic state consisting of a settled past together with the constraints that currently govern the space of admissible future continuations.*

Remark 12. *Configurations do not contain values, propositions, or truth assignments. They encode only history and constraint.*

B.2 Small-Step Transition Relation

We define a small-step transition relation

$$(h, \Sigma) \longrightarrow (h', \Sigma').$$

Each transition rule satisfies two invariants:

1. *Prefix preservation*: $h \preceq h'$,
2. *Constraint monotonicity*: admissible futures do not expand.

The transition relation formalizes execution as progressive restriction of future possibility.

B.3 Primitive Rewrite Rules

We define the following canonical rule schemas.

B.3.1 POP (Elimination)

The POP rule removes futures violating a constraint.

$$\frac{h' \in \text{Fut}(h) \quad h' \notin \sigma(h)}{(h, \{\sigma\} \cup \Sigma) \longrightarrow (h, \Sigma)}$$

This rule enforces the constraint without altering history. It eliminates incompatible continuations but introduces no new structure.

B.3.2 BIND (Precedence Constraint)

The BIND rule introduces ordering constraints on future events.

$$(h, \{\text{BIND}(e_i \prec e_j)\} \cup \Sigma) \longrightarrow (h, \Sigma')$$

where Σ' contains an operator enforcing the precedence relation $e_i \prec e_j$ on all future extensions. BIND does not settle events; it constrains their admissible orderings.

B.3.3 MERGE (Identification)

The MERGE rule identifies compatible histories.

$$(h_1, h_2, \text{MERGE}) \longrightarrow (h, \Sigma)$$

where h is the least upper bound of h_1 and h_2 under prefix order, if such a history exists. MERGE fails when no consistent joint history exists, producing deadlock.

B.3.4 COLLAPSE (Forgetting)

The COLLAPSE rule erases internal distinctions while preserving admissibility.

$$(h, \text{COLLAPSE}) \longrightarrow (h', \Sigma)$$

where h' is a quotient of h under an equivalence relation that preserves future availability. COLLAPSE reduces historical resolution without increasing possibility.

B.4 Invariant Preservation

[Prefix Preservation] For any transition $(h, \Sigma) \rightarrow (h', \Sigma')$, we have

$$h \preceq h'.$$

Proof. POP, BIND, and COLLAPSE do not extend history. MERGE produces a history that contains both input histories as prefixes. No rule removes past events. $\square$

B.5 Soundness Theorem

[Soundness] If

$$(h, \Sigma) \longrightarrow (h', \Sigma'),$$

then

$$\text{Fut}(h') \subseteq \text{Fut}(h).$$

Proof. The claim follows by inspection of the rewrite rules. In the case of POP, admissible futures are restricted by intersection with the constraint set $\sigma(h)$. The BIND rule excludes precisely those future histories that violate the imposed precedence relations. The MERGE rule retains only those

histories that are jointly consistent with the merged inputs, while the COLLAPSE rule preserves admissible futures up to an explicitly defined equivalence relation. In all cases, the effect of rewriting is either to preserve or to eliminate elements of $\text{Fut}(h)$; no rule introduces histories that were not already admissible. It follows that admissible futures are preserved or strictly reduced under every transition. $\square$

Soundness therefore guarantees that semantic execution respects the irreversibility of event histories.

B.6 Completeness Theorem

[Completeness] Let $C \subseteq \text{Fut}(h)$ be any constraint expressible as a finite intersection of operator images. Then there exists a finite sequence of rewrite steps yielding a configuration whose admissible futures are exactly C .

Proof. By assumption,

$$C = \bigcap_{i=1}^n \sigma_i(h).$$

Applying the corresponding operators via POP in any order yields the desired restriction. Order is irrelevant due to commutativity and associativity of intersection. $\square$

Completeness ensures that the operational semantics can realize all expressible constraints.

B.7 Deadlock and Collapse

Definition 33 (Semantic Deadlock). *A configuration (h, Σ) is in deadlock if*

$$\bigcap_{\sigma \in \Sigma} \sigma(h) = \emptyset.$$

[Deadlock Persistence] Once deadlock is reached, no further transitions are possible.

Proof. All rewrite rules presuppose non-empty admissible futures. When none exist, no rule applies. $\square$

Deadlock corresponds to semantic absolutism: no continuation is permitted.

B.8 Determinacy and Confluence

Proposition 8 (Confluence). *Operator application is confluent: the order of operator enforcement does not affect the resulting admissible future space.*

Proof. Operator composition is set intersection, which is associative and commutative. All reduction paths therefore converge to the same admissible future space. $\square$

Confluence guarantees determinacy of semantic outcome.

B.9 Appendix Summary

This appendix has developed a small-step operational semantics for the constraint-first framework and demonstrated that semantic execution preserves the irreversibility invariants imposed by event histories. It has established soundness and completeness results for the execution model, showing that all and only admissible future restrictions can be realized through finite sequences of rewrite steps. In addition, the appendix has provided a formal characterization of semantic deadlock in terms of empty future spaces and shown that operator application is confluent, ensuring that the order of constraint enforcement does not affect the resulting admissible futures.

These results justify treating meaning as executable constraint rather than denotation and provide the technical backbone for the categorical semantics developed elsewhere.

C Functorial Semantics: Expanded Constructions and Proofs

This appendix lifts the operational semantics developed in Appendix B to a categorical level. The aim is to demonstrate that constraint-first meaning is not merely an execution model but a structural semantics with well-defined invariants, equivalences, and transformation laws. The categorical perspective clarifies which aspects of meaning are preserved under reinterpretation and why certain semantic failures correspond to formal type errors.

C.1 The Category of Histories

Definition 34 (Category **Hist**). *Define **Hist** as the category whose objects are event histories $h \in \mathcal{H}$ and whose morphisms are given by prefix inclusion: there exists a unique morphism $h \rightarrow h'$ if and only if $h \preceq h'$.*

[Thinness] The category **Hist** is thin: between any two objects there exists at most one morphism.

Proof. Prefix order is antisymmetric and transitive. Thus for any $h, h' \in \mathcal{H}$, the relation $h \preceq h'$ uniquely determines the existence of a morphism, and no parallel morphisms are possible. $\square$

Thinness reflects the irreversibility of time: histories extend but do not branch backward.

C.2 The Category of Constraint Spaces

Definition 35 (Constraint Space). *A constraint space is a subset $C \subseteq \mathcal{H}$ such that whenever $h \in C$ and $h \preceq h'$, then $h' \in C$.*

Constraint spaces are upward closed under prefix order and represent sets of admissible futures.

Definition 36 (Category **Constr**). *Define **Constr** as the category whose objects are constraint spaces and whose morphisms are monotone functions preserving inclusion.*

The category **Constr** forms a complete lattice under intersection and union.

Proof. Upward closure is preserved under arbitrary intersections and unions. Hence the set of constraint spaces is closed under these operations, yielding a complete lattice structure. $\square$

This lattice structure reflects the combinatorial nature of constraint composition.

C.3 The Meaning Functor

Definition 37 (Meaning Functor). *Define a functor*

$$\mathcal{M} : \mathbf{Hist} \rightarrow \mathbf{Constr}$$

by

$$\mathcal{M}(h) = \text{Fut}(h),$$

and for a morphism $h \rightarrow h'$,

$$\mathcal{M}(h \rightarrow h')(C) = C \cap \text{Fut}(h').$$

[Functoriality] The mapping $\mathcal{M}$ is a well-defined functor.

Proof. Identity: for any history h ,

$$\mathcal{M}(\text{id}_h)(C) = C \cap \text{Fut}(h) = C,$$

since $C \subseteq \text{Fut}(h)$ by definition.

Composition: if $h \preceq h' \preceq h''$, then

$$\mathcal{M}(h \rightarrow h'') = \mathcal{M}(h' \rightarrow h'') \circ \mathcal{M}(h \rightarrow h'),$$

because intersection with $\text{Fut}(h'')$ factors through intersection with $\text{Fut}(h')$. $\square$

The meaning functor assigns to each history its space of admissible continuations.

C.4 Operators as Endofunctors

Definition 38 (Operator Endofunctor). *Each operator σ induces an endofunctor*

$$F_\sigma : \mathbf{Constr} \rightarrow \mathbf{Constr}$$

defined by

$$F_\sigma(C) = C \cap \sigma(h),$$

where h is any history generating the constraint space C .

[Well-Definedness] The definition of F_σ does not depend on the choice of representative history h .

Proof. If $\text{Fut}(h) = \text{Fut}(h')$, then h and h' are equivalent under prefix closure. By operator monotonicity, $\sigma(h) = \sigma(h')$, so the intersection defining $F_\sigma(C)$ is invariant. $\square$

Thus operators act uniformly on constraint spaces.

C.5 Composition and Identity

Proposition 9. *Operator composition corresponds to functor composition:*

$$F_{\sigma_1 \circ \sigma_2} = F_{\sigma_1} \circ F_{\sigma_2}.$$

Proof. Both sides evaluate to intersection with $\sigma_1(h) \cap \sigma_2(h)$ for any representative history h . $\square$

The identity operator induces the identity endofunctor on $\mathbf{Constr}$.

C.6 Reinterpretation as Natural Transformation

Definition 39 (Reinterpretation). *A reinterpretation of a symbol is a natural transformation*

$$\eta : F_\sigma \Rightarrow F_{\sigma'}.$$

[Naturality Condition] For all morphisms $f : C \rightarrow C'$ in **Constr**,

$$\eta_{C'} \circ F_\sigma(f) = F_{\sigma'}(f) \circ \eta_C.$$

Failure of this condition corresponds to semantic drift: reinterpretation that depends on context rather than preserving structural meaning.

C.7 Identity and Fixed Points

Definition 40 (Stable Identity). *A constraint space C is a stable identity if and only if*

$$F_\sigma(C) = C.$$

Proposition 10. *Stable identities are precisely the fixed points of the operator endofunctor system.*

Proof. This follows immediately from the definition of fixed points in **Constr**. $\square$

Identities are thus not primitive objects but stabilized regions of constraint space.

C.8 Reification as a Categorical Error

Definition 41. *Reification is the attempt to treat an endofunctor or a natural transformation as an object of **Hist**.*

Such an attempt destroys functorial structure.

Proof. Objects of **Hist** are histories, not transformations. Treating endofunctors or natural transformations as objects eliminates the distinction between objects and morphisms, collapsing composition and violating naturality. $\square$

This formalizes reification as a categorical type error.

C.9 Appendix Summary

This appendix has established the categorical structure underlying the framework, including the categories of event histories and constraint spaces and the functor that maps histories to their admissible futures. Within this setting, symbolic operators were shown to correspond to endofunctors on constraint spaces, while reinterpretation was formalized as natural transformation. Identity was characterized as a fixed point of constraint composition, and reification was identified as a categorical type error arising from the misclassification of morphism-level structure as object-level content.

Taken together with the operational results developed earlier, these constructions demonstrate that constraint-first meaning admits both an executable semantics and an abstract categorical characterization. Crucially, the same structural invariants are preserved across these levels of description, ensuring that the formalism remains coherent under both execution and abstraction.

D Notation Glossary and Meta-Theoretic Remarks

This appendix consolidates the notation used throughout the paper and records meta-theoretic properties of the formal system. Its purpose is expository and clarificatory rather than developmental. No new formal machinery is introduced; instead, this appendix makes explicit the commitments, limits, and interpretive status of the framework as a whole.

D.1 Notation Glossary

Symbol	Meaning
$\mathcal{H}$	Set of event histories
h, h'	Individual event histories
e	An atomic irreversible event
$\prec$	Precedence relation on events
$\preceq$	Prefix order on histories
$\text{Fut}(h)$	Admissible futures of history h
$\mathcal{O}$	Semantic type of operators
$\mathcal{R}$	Semantic type of referential terms
σ, τ	Operators (future-constraint functions)
Σ	Finite multiset of active operators
(h, Σ)	Operational configuration
Hist	Category of event histories
Constr	Category of constraint spaces
$\mathcal{M}$	Meaning functor Hist $\rightarrow$ Constr
F_σ	Endofunctor induced by operator σ
η	Natural transformation (reinterpretation)
$\circ$	Operator or functor composition
$\emptyset$	Empty future space (semantic deadlock)

Remark 13. *All symbols are typed implicitly by context. Ambiguity of semantic type is treated as a formal error rather than an interpretive ambiguity.*

D.2 Meta-Theoretic Status of the System

This subsection records what has been established formally and what remains outside the scope of proof.

Consistency. The system is consistent relative to standard set theory. No rule introduces contradiction because no rule asserts propositions. All semantics is structural, monotone, and eliminative rather than generative.

Soundness and Completeness. Soundness and completeness were proven with respect to the operational semantics in Appendix B. These results are internal: they concern faithful execution of constraint restriction rather than correspondence to an external model of truth.

Decidability. Decidability of non-empty future space depends on the class of operators admitted. For finite operator sets with computable constraint predicates, deadlock detection is decidable. In the general case, undecidability is expected.

Expressive Power. The system is strictly more expressive than propositional semantics, but strictly less expressive than unrestricted higher-order logic. Its expressive strength derives from irreversibility and constraint composition rather than from quantification over truth values.

D.3 Negative Results and Non-Claims

For the avoidance of misunderstanding, it is important to state explicitly what the framework does not claim. It does not assert that all disagreement is semantic in origin, nor does it suggest that referential semantics can be eliminated or replaced wholesale. It does not maintain that operator-level interpretation is sufficient to resolve every form of conflict, and it does not propose that formal models of constraint provide an exhaustive account of lived experience, social dynamics, or psychological motivation.

These negative claims are not incidental qualifications. They are structural boundaries that prevent theoretical overreach and clarify the specific explanatory target of the framework. By delineating what lies outside its scope, the analysis preserves both its neutrality and its analytical force.

Remark 14. *The framework diagnoses a specific semantic failure mode. It is not a universal solvent.*

D.4 Formalism and Interpretation

The formal system constrains admissible interpretations but does not determine them uniquely. Multiple narratives, institutions, or practices may instantiate the same constraint topology.

This explains how semantic equivalence can coexist with surface disagreement.

Proposition 11. *If two symbolic systems induce naturally isomorphic meaning functors, then they are semantically equivalent with respect to admissible futures.*

Proof. Natural isomorphism preserves all structural relations in **Constr**, including inclusion, intersection, and fixed points. Hence the induced spaces of admissible futures coincide. $\square$

D.5 Reification at the Meta-Level

Reification can itself be understood meta-theoretically as a collapse of typing discipline.

Definition 42 (Meta-Reification). *Meta-reification occurs when the semantic framework itself is treated as a descriptive ontology rather than as a constraint model.*

This appendix explicitly avoids that error. The formalism does not describe what exists; it specifies how meaning constrains continuation.

D.6 Appendix Summary

This appendix has served to consolidate the formal presentation of the framework by fixing notation unambiguously and recording the meta-theoretic status of the system. It has clarified which structural guarantees are established by the formalism and which questions are intentionally left open, thereby preventing overextension or misinterpretation of the results. In doing so, it has made explicit the scope conditions under which the framework is intended to apply and the limits beyond which its claims do not extend.

In addition, the appendix has articulated criteria for semantic equivalence in terms of preserved constraint structure, rather than surface representation or interpretive narrative. Finally, it has introduced a principled safeguard against reification at the level of the theory itself, ensuring that the formal apparatus is understood as a model of constraint and coordination rather than as a descriptive ontology.

Taken together, these clarifications complete the formal specification of the constraint-first semantic framework and establish the conditions under which its results should be interpreted.

References

- J. L. Austin. *How to Do Things with Words*. Oxford University Press, 1962.
- Robert Brandom. *Making It Explicit*. Harvard University Press, 1994.
- Keith Donnellan. Reference and definite descriptions. *The Philosophical Review*, 1966.
- Gottlob Frege. *On Sense and Reference*. Zeitschrift für Philosophie und philosophische Kritik, 1892.
- H. L. A. Hart. *The Concept of Law*. Oxford University Press, 1961.
- Saul Kripke. *Naming and Necessity*. Harvard University Press, 1980.
- Bertrand Russell. *On Denoting*. Mind, 1905.
- John Searle. *Speech Acts*. Cambridge University Press, 1969.
- Ludwig Wittgenstein. *Philosophical Investigations*. Blackwell, 1953.